Cambridge (CIE) A Level Computer Science

Your notes

Program Design

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Structure charts

What is a structure chart?

- A structure chart is a **modelling tool** used in the **design stage** of program development
- It helps to **decompose a problem** into smaller, manageable **sub-tasks**, representing each as a **module**
- Structure charts focus on **what the program does**, not how it does it

Example

- A program that asks a user to enter their username and password
- Checks if the password matches the stored password and uses a counter to track how many times it is checked
- If the password matches, a successful message is displayed, else after three failed attempts a 'denied' message is displayed
- A structure chart might look like:

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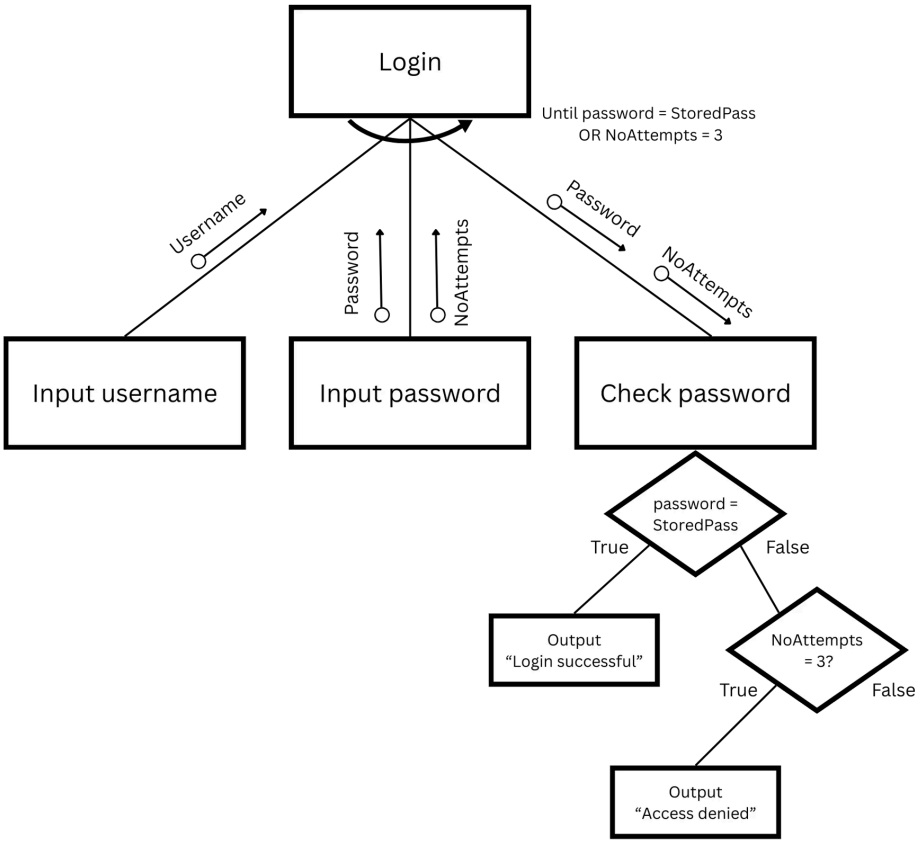

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Key features

Feature	Explanation
Top-down design	The chart starts with the **main program** and breaks it into sub-modules
Decomposition	Each module represents a **specific task** or **function**
Module boxes	Each module is shown as a **box**
Vertical lines	Show **control flow** – which module calls which
Parameter arrows	**Arrows** pointing into modules show **data being passed in or returned**

Stepwise refinement	Each level adds **more detail** to the task above
Diamond shape	Shows a **condition** that could be **true or false**
Semi-circular arrow	Indicates **repetition**

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Pseudocode from structure chart

- The first step is to create an **identifier table**

Identifier	Data type	Description
Username	STRING	Stores the username entered by the user
Password	STRING	Stores the password entered by the user
StoredPass	STRING	The correct password stored in the system for comparison
NoAttempts	INTEGER	Counts the number of failed login attempts
LoginSuccess	BOOLEAN	Indicates whether the login was successful

- Next, identify is any **functions** or **procedures** could be used

Module	Type	Purpose
GetCredentials	**PROCEDURE**	Asks the user to input username and password
CheckPassword	**FUNCTION**	Compares input password with stored password and returns TRUE/FALSE
ShowAccessMessage	**PROCEDURE**	Displays success or failure message

- Any finally, write the pseudocode

```
DECLARE Username : STRING
DECLARE Password : STRING
DECLARE StoredPass : STRING
DECLARE NoAttempts : INTEGER
DECLARE LoginSuccess : BOOLEAN

StoredPass ← "admin123"
NoAttempts ← 0
LoginSuccess ← FALSE
```

```
PROCEDURE GetCredentials()
   OUTPUT "Enter username: "
   INPUT Username
   OUTPUT "Enter password: "
   INPUT Password
ENDPROCEDURE

FUNCTION CheckPassword(P : STRING) RETURNS BOOLEAN
   IF P = StoredPass THEN
      RETURN TRUE
   ELSE
      RETURN FALSE
   ENDIF
ENDFUNCTION

PROCEDURE ShowAccessMessage(Success : BOOLEAN)
   IF Success = TRUE THEN
      OUTPUT "Login successful"
   ELSE
      OUTPUT "Access denied"
   ENDIF
ENDPROCEDURE

WHILE NoAttempts < 3 AND LoginSuccess = FALSE
   CALL GetCredentials()
   LoginSuccess ← CheckPassword(Password)

   IF LoginSuccess = FALSE THEN
      OUTPUT "Incorrect password"
      NoAttempts ← NoAttempts + 1
   ENDIF
ENDWHILE

CALL ShowAccessMessage(LoginSuccess)
```

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